

Producer ↔ consumer loop through the affordance node

PRODUCER
table_concept

create / refresh the node; keep
the parent feedback attrs fresh

DDS-replicated

```
aff_policy = servo
aff_goal_attrs | ops
| values
aff_stable_n .
timeout_ms . on_fail
```

```
write active=0, pending=0 — closes
the loop
```

```
active=claim
pending=done
```

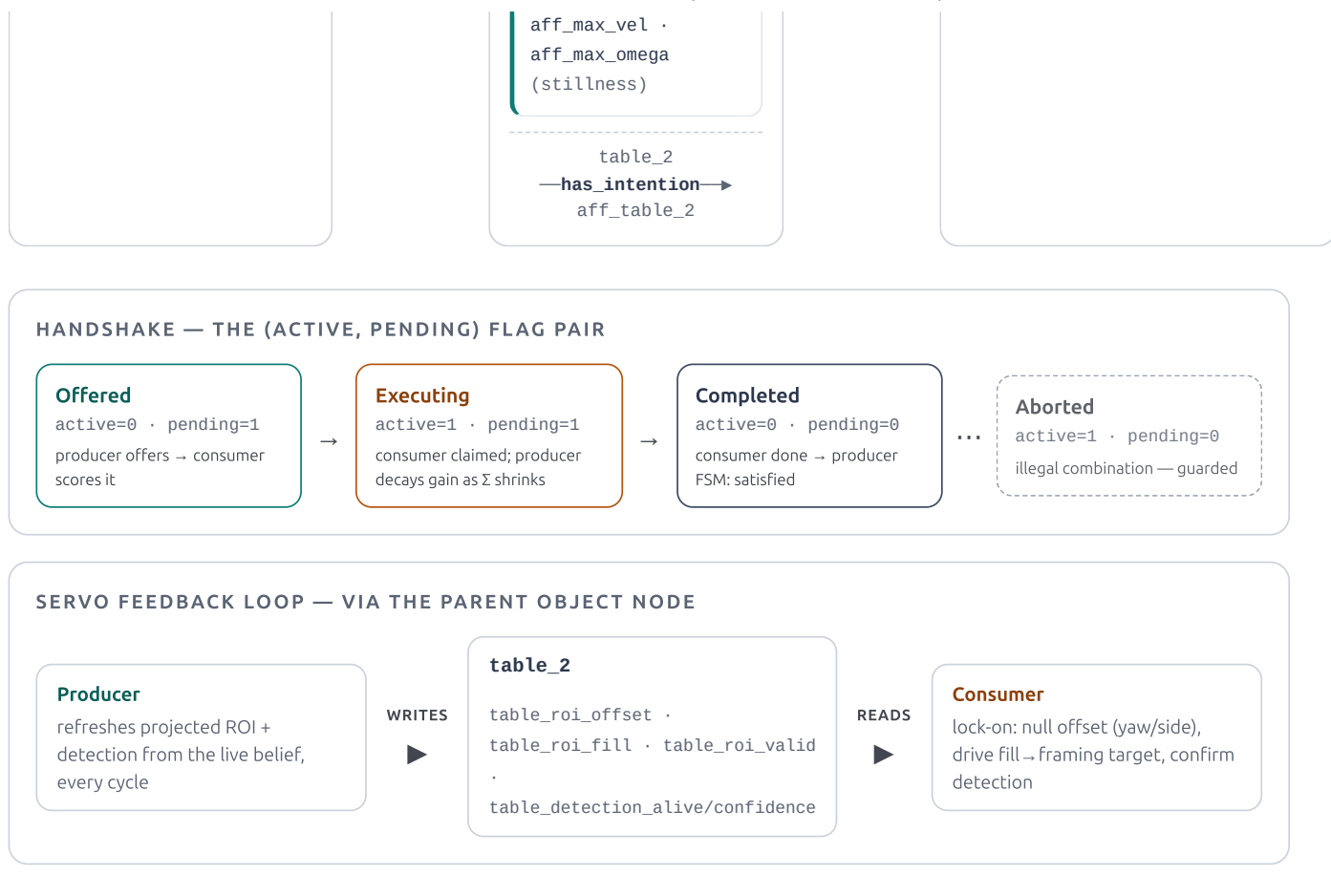


Figure 1. Affordance as a shared-graph contract between an active-inference perception agent and a controller. The producer's next-best-view planner scores candidate faces by the D-optimal expected entropy reduction on its belief covariance, bounded by the adequacy gap to a consumer-declared precision Σ^* , and publishes an **object-relative** viewpoint constraint (ranked faces + per-face gain in nats + stand-off band + framing + Σ^*) rather than a world pose. The controller selects among all competing affordances by expected free energy $G = \lambda \cdot \text{nav_dist} - \text{epistemic_gain}$, claims one through the **active/pending** handshake, resolves the constraint into a collision-free reachable pose (Part B), and executes it via a servo feedback loop bound to the parent object node. As evidence accumulates, the belief covariance contracts, the adequacy gap and thus **epistemic_gain** fall to zero, and the affordance self-extinguishes — a threshold-free "done" set by the consumer's precision demand.